Aditya Chilla

USA | +1 585-410-8788 | adityacv2000@gmail.com | LinkedIn | GitHub | Portfolio

Summary

Engineer with 2+ years of experience developing ML pipelines, AI models, and containerized software deployments on AWS using Python, SQL, PyTorch, XGBoost, GoLang, Git, CI/CD and LangChain in the biotech, utilities, and healthcare industries.

Experience

Fownd Inc — Dec 2025 – Present
Software Engineer — *Remote*

- Implemented an automated load testing scripts simulating up to 200 concurrent users, enabling infrastructure improvements to AWS cloud performance, data processing pipelines, and scalability while debugging critical bottlenecks for each AWS service(ECS, EC2, Bedrock Services, etc..).
- Developed a microservice-based payment system using Go, Docker, Cassandra, and Redis, supporting secure and scalable enterprise transaction processing.
- Architected a scalable observability stack (Grafana, Prometheus, Loki, AWS X-Ray) on a dedicated AWS EKS cluster using Terraform, enabling end-to-end production-level backend/ API monitoring, tracing, and logging for real-time problem detection and diagnosis.

Freelancer – VigVen — Sep 2025 – Nov 2025
Data & Software Engineer — *Remote*

- Utilised Python for building an automation tool with a PyQt GUI that collects and structures test execution data, integrating it into a database to support model evaluation workflows.
- Built a Docker and MSSQL-based automated data pipeline to store and analyze test results, integrating model deployment processes for reproducible workflows and future analysis.

ChaseTech Solz — Mar 2024 – Jul 2025
Data Scientist — *Philadelphia, PA, USA*

- Developed ML models using Python, PySpark and SQL for patient enrollment assessment and clinical trial site performance forecasting across 50+ sites in 20+ countries, processing 50,000+ patient records with to enable timely site selection and enrollment rate decision making for a biotech client.
- Designed interactive PowerBI dashboards to monitor clinical KPIs across 50+ sites, enabling an informed decision-making and reducing time-to-insight for senior stakeholders.

Avangrid – Iberdrola Group — Sep 2023 – Dec 2023
Data Scientist — *Rochester, NY, USA*

- Assessed utility pole damage patterns caused by osprey nests, using SQL, Python, feature engineering, and Databricks to develop a classification model, proactively generated insights to inform operational decision-making beyond the initial project scope.
- Engineered an interactive dashboard to visualize power outage trends, confidently communicating findings to both technical colleagues and non-technical stakeholders, enhancing monitoring efficiency by 30%.

University of Rochester Medical Center – Dept. of Pediatrics — May 2023 – Sep 2023
Graduate Research Assistant — *Rochester, NY, USA*

- Analyzed rat retina imaging data using OpenCV and statistical analysis, exploring and interpreting raw data to surface insights and automate early-stage detection of vascular patterns, supporting research on Retinopathy of Prematurity.
- Trained a Siamese neural network model using PyTorch to classify vascular patterns in mouse retina images, achieving 82% accuracy and significantly reducing manual image review time; translated complex model outputs into clear, accessible findings for senior research audiences.

Projects

Automated Pre-Match Opponent Analysis Platform | *Python, XGBoost, K-Means, Llama 3, mplsoccer*

- Built an end-to-end opponent profiling system using raw StatsBomb open data, applying XGBoost xG modelling, K-Means tactical clustering, and LLM-powered (Llama 3 via Ollama) narrative generation to automate executive pre-match coaching reports with mplsoccer pitch visualisations.

Detection of Shilling Attacks in Recommender Systems | *Python, Random Forest, scikit-learn*

- Led a four-member team to develop and optimise a Random Forest classifier in Python for detecting shilling attacks in recommender systems, incorporating RDMA and DegSim features to achieve 93% accuracy and publishing findings in a peer-reviewed journal.

Education

University of Rochester — Rochester, NY, USA
Master of Science, Data Science (GPA: 3.5/4.0) — *Sep 2022 – Dec 2023*

Gandhi Institute of Technology and Management (GITAM) — Visakhapatnam, India
Bachelor of Technology, Computer Science (GPA: 8.51/10) — *Jun 2018 – Apr 2022*

Skills

Programming: Python, C++, Java, C, SQL, Linux/Unix, GoLang, Terraform
AI & ML: Deep Learning, LLMs, NLP, Time Series Forecasting, Predictive Modelling, A/B Testing, Hypothesis Testing
Frameworks & Tools: TensorFlow, PyTorch, XGBoost, scikit-learn, MLflow, OpenCV, LangChain, Ollama
Data & Cloud: AWS, GCP, Databricks, PySpark, Hadoop, Docker, Kubernetes, Terraform, Helm
Visualisation: PowerBI, Tableau, Matplotlib, Seaborn, mplsoccer
Practices: MLOps, Model Deployment, Model Evaluation, Data Processing, Agile